Probability and Independence

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Overview

Conditional probability

Definition: The conditional probability of event A given event B is:

$$\Pr[A|B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

Conceptually, if we limit ourselves to the outcomes in B, how likely is an outcome in A?

Example:

- A: Die shows a number divisible by 3. $\Pr[A] = 1/3$. (3 and 6 from the six possibilities.)
- B: Die shows an odd number. $\Pr[B] = 1/2$.
- What does $\Pr[A \cap B]$ mean? Die shows an odd number divisible by 3. $\Pr[A \cap B] = 1/6$ (only 3).
- What does $\Pr[A|B]$ mean? Die shows a number divisible by 3 given that it's odd. $\Pr[A|B] = 1/3$ (probability of picking 3 from 1, 3, 5). Also, $\frac{1}{6}/\frac{1}{2} = \frac{1}{6} \times 2 = \frac{1}{3}$.

Independence

Definition: Event A is *independent of* event B iff

$$Pr[A|B] = \Pr[A].$$

If $\Pr[B] = 0$, we say it is independent of any other event including itself.

Example:

- A: Die shows the maximum or minimum number. $\Pr[A] = 1/3$. (1, 6 from the six possibilities.)
- B: Die shows an odd number. $\Pr[B] = 1/2$.
- $\Pr[A|B] = 1/3$. (1 from 1,3,5.) So, A and B are independent.
- C: Die shows an even number. $\Pr[C] = 1/2$.
- Are B and C independent? No, $\Pr[C|B] = 0 \neq \Pr[C]$. Common misconception. Independent does not mean disjoint.

Independent events multiply

Theorem: A is independent of B iff

$$\Pr[A \cap B] = \Pr[A] \cdot \Pr[B].$$

Proof: By cases.

- Case 1: If $\Pr[A] = 0$ or $\Pr[B] = 0$, then $\Pr[A \cap B] = 0$. Equality and independence are both achieved.
- Case 2: Otherwise, A is independent of B iff $\Pr[A|B] = \Pr[A]$ by definition. Substituting in the definition of conditional probability, we have $\Pr[A|B] = \Pr[A]$ iff $\Pr[A \cap B]/Pr[B] = Pr[A]$. Multiplying both sides by $\Pr[B]$, we have $\Pr[A \cap B]/Pr[B] = Pr[A]$ iff $\Pr[A \cap B] = Pr[A] \cdot Pr[B]$. QED.

Mutual Independence

Definition: A set of events $E_1, E_2, \ldots, E_n$ is *mutually independent* iff for all subsets $S \subseteq [1, n]$,

$$\Pr\left[\bigcap_{j \in S} E_j\right] = \prod_{j \in S} \Pr[E_j].$$

Example: If we toss n fair coins, the tosses are mutually independent iff for every subset of m coins, the probability that every coin in the subset comes up heads is 2^{-m}.

Independent missions

Mars	one-time	person	.028
Mars	one-time	robot	.042
Mars	reusable	person	.252
Mars	reusable	robot	.378
Moon	one-time	person	.012
Moon	one-time	robot	.018
Moon	reusable	person	.108
Moon	reusable	robot	.162

$\Pr[\text{Mars}] = 0.028 + 0.042 + 0.252 + 0.378 = 0.7$

$\Pr[\text{one-time}] = 0.028 + 0.042 + 0.012 + 0.018 = 0.1$

$\Pr[\text{person}] = 0.028 + 0.252 + 0.012 + 0.108 = 0.4$

$\Pr[\text{Mars AND one-time}] = 0.028 + 0.042 = 0.07 = 0.7 \times 0.1$

$\Pr[\text{Mars AND person}] = 0.028 + 0.252 = 0.28 = 0.7 \times 0.4$

$\Pr[\text{one-time AND person}] = 0.028 + 0.012 = 0.04 = 0.1 \times 0.4$

$\Pr[\text{Mars AND one-time AND person}] = 0.028 = 0.7 \times 0.1 \times 0.4$

Pairwise independence isn't mutual independence

If A is independent of B and C, and B and C are independent of each other, how could A, B, and C not be independent??

Example: Morley, Kyran, and Will each pick a bit 0/1 uniformly at random.

- A: Morley + Kyran $\equiv 1 \pmod 2$
- B: Morley + Will $\equiv 1 \pmod 2$
- C: Kyran + Will $\equiv 1 \pmod 2$

Claim 1: These events are all pairwise independent.

For example, $\Pr[A] = 1/2$. $\Pr[A|B] = \frac{1}{4}/\frac{1}{2} = 1/2$.

Claim 2: These events are not mutually independent.

$\Pr[A \cap B \cap C] = 0$. Not $1/8$!

k-wise does not imply $(k+1)$-wise mutual independence.